LECTURE NOTES: GALOIS THEORY IN ALGEBRA AND TOPOLOGY

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1. Introduction

The aim of these notes is to explain, using basic complex topology, how covering spaces of Riemann surfaces control certain Galois groups, and to apply this to the absolute Galois group of $\mathbb{C}(t)$.

In §2 we recall some facts from the previous lectures.

In §3 we fix a connected compact Riemann surface X and a subset $X' \subset X$ with finite complement. We define the field $K_{X'}$ as the composition of all finite extensions of $\mathcal{M}(X)$ coming from covers of X that are unramified over X'. We then prove that $K_{X'} \mid \mathcal{M}(X)$ is Galois and identify its Galois group with the profinite completion of the topological fundamental group of X'.

In §4 we specialize to $X = \mathbb{P}^1(\mathbb{C})$, so that $\mathcal{M}(X) \cong \mathbb{C}(t)$. We state and prove Douady's theorem: the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{C}(t)} \mid \mathbb{C}(t))$ is the free profinite group on the set $\mathbb{C}$.

2. Two fundamental equivalences

Before our discussion we recall two results from the previous courses. These results will be used repeatedly throughout the course.

Theorem 2.1. *Let X be a connected compact Riemann surface. Then there is an equivalence of categories*

$$\mathrm{Riem}_X^{\mathrm{fin}} \ \simeq \ \mathrm{F\acute{E}t}_{\mathcal{M}(X)}^{\mathrm{opp}}$$

where:

- $\mathrm{Riem}_X^{\mathrm{fin}}$ *denotes the category of compact Riemann surfaces Y mapping holomorphically onto X;*
- $\mathrm{F\acute{E}t}_{\mathcal{M}(X)}^{\mathrm{opp}}$ *denotes the opposite category of finite étale $\mathcal{M}(X)$-algebras.*

Under this equivalence, finite Galois branched covers of X correspond to finite field extensions of $\mathcal{M}(X)$ of the same degree.

Theorem 2.2 (Riemann existence theorem with punctures)**.** *Let X be a connected compact Riemann surface and let $S \subset X$ be a finite set. Set $X' := X \setminus S$. Then there is an equivalence of categories*

$$\mathrm{Riem}_{X,S}^{\mathrm{fin}} \ \simeq \ \mathrm{Cov(X - S)}$$

where $\mathrm{Riem}_{X,S}^{\mathrm{fin}}$ is the full subcategory consisting of $Y \to X$ with all branching points lying in S, and $\mathrm{Cov}(X - S)$ is the category of finite topological covers of $X - S$.

3. Covers of Riemann surfaces and Galois groups

Let X be a connected compact Riemann surface and let $X' \subset X$ be a subset so that $S = X \setminus X'$ is a finite set. Write $\mathcal{M}(X)$ for the field of meromorphic functions on X, and fix an algebraic closure $\overline{\mathcal{M}(X)}$ of $\mathcal{M}(X)$.

Definition 3.1. A finite extension $L \mid \mathcal{M}(X)$ *comes from a (compact) Riemann surface* if there exists a connected compact Riemann surface Y and a nonconstant holomorphic map $p : Y \to X$ such that $\mathcal{M}(Y) \cong L$ as extensions of $\mathcal{M}(X)$.

We say that such a map $p : Y \to X$ *restricts to a cover over* X' if the induced map $p^{-1}(X') \to X'$ is a topological cover (or equivalently: p is unramified above X').

Definition 3.2. Let $K_{X'} \subset \overline{\mathcal{M}(X)}$ be the composition of all finite subextensions $L \mid \mathcal{M}(X)$ that come from a connected compact Riemann surface Y and restrict to a cover over X'.

Lemma 3.3. *Let $L \mid \mathcal{M}(X)$ be a finite subextension of $K_{X'} \mid \mathcal{M}(X)$. Then there exists a connected compact Riemann surface Y and a holomorphic map $p : Y \to X$ such that $\mathcal{M}(Y) \cong L$ and p restricts to a cover over X'.*

Proof. By definition, $K_{X'}$ is the composition inside a fixed algebraic closure of $M(X)$ of all finite extensions $L_i \mid M(X)$ that come from connected compact Riemann surfaces and restrict to a cover over $X^{'}$. Thus every finite subextension $L \mid M(X)$ of $K_{X'}$ is contained in a composition $L_1 \cdots L_m$ of finitely many such fields. It is therefore enough to show that the property

> *"comes from a connected compact Riemann surface and restricts to a cover over $X^{'}$"*

is stable under taking compositions and under taking intermediate subextensions.

Step 1: stability under compositions. Let $L_1 \mid M(X)$ and $L_2 \mid M(X)$ be finite extensions coming from holomorphic maps $p_1 : Y_1 \to X$ and $p_2 : Y_2 \to X$ of connected compact Riemann surfaces, each restricting to a cover over $X^{'}$. Consider the fibre product $Y_1 \times_X Y_2$ and let Y be a connected component of it. Then the induced map $p : Y \to X$ is holomorphic, and since covers are stable under base change, the restriction $p^{-1}(X^{'}) \to X^{'}$ is again a (finite) topological cover.

On the level of function fields, the anti-equivalence between compact Riemann surfaces over X and finite extensions of $M(X)$ identifies connected components of $Y_1 \times_X Y_2$ with field factors of $L_1 \otimes_{M(X)} L_2$. More precisely, if α is a primitive element of $L_1 \mid M(X)$ with minimal polynomial $f \in M(X)[T]$, then the factorization of f into irreducibles over L_2 corresponds to the decomposition of $Y_1 \times_X Y_2$ into connected components, and the component corresponding to the factor determined by the fixed embeddings of L_1 and L_2 into $\overline{M(X)}$ has function field equal to the composition $L_1 L_2$. Hence $L_1 L_2 \mid M(X)$ comes from a connected compact Riemann surface and restricts to a cover over $X^{'}$.

Iterating, we see that $K_{X'}$ is the union of a tower of finite subextensions

$$L_1 \subset L_1 L_2 \subset L_1 L_2 L_3 \subset \cdots$$

each coming from a connected compact Riemann surface restricting to a cover over $X^{'}$.

Step 2: stability under intermediate subextensions. Let $L \mid M(X)$ be a finite extension with $M(X) \subset K \subset L$. Assume that L comes from a holomorphic map $p : Y \to X$ of connected compact Riemann surfaces restricting to a cover over $X^{'}$. Let Z be the compact Riemann

surface corresponding to the intermediate field K (and write $q : Z \to X$ for the induced holomorphic map), so that we have a factorization

$$Y \xrightarrow{r} Z \xrightarrow{q} X$$

with $\mathcal{M}(Y) \cong L$ and $\mathcal{M}(Z) \cong K$.

Fix a point $x \in X^{'}$. Since p restricts to a cover over $X^{'}$, every point of $X^{'}$ has exactly $[L : M(X)]$ preimages in Y (counted without multiplicity). On the other hand, for any point of X the fibre of q has at most $[K : M(X)]$ points, and for any point $z \in Z$ the fibre of r has at most $[L : K]$ points. Therefore, for $x \in X^{'}$ we have the chain of inequalities

$$\begin{aligned}[L : M(X)] = |p^{-1}(x)| &= \sum_{z \in q^{-1}(x)} |r^{-1}(z)| \\ &\leq \sum_{z \in q^{-1}(x)} [L : K] = [L : K] \cdot |q^{-1}(x)| \\ &\leq [L : K] \cdot [K : M(X)] = [L : M(X)].\end{aligned}$$

All inequalities are thus equalities. In particular, for every $x \in X^{'}$ we get $|q^{-1}(x)| = [K : M(X)]$, which means that q has constant fibre cardinality on $X^{'}$. Equivalently, q is unramified above $X^{'}$, i.e. q restricts to a topological cover over $X^{'}$.

Combining Steps 1 and 2, every finite subextension L of $K_{X'} \mid \mathcal{M}(X)$ comes from a connected compact Riemann surface and restricts to a cover over $X^{'}$. □

Theorem 3.4. *Let X be a connected compact Riemann surface and let $X' \subset X$ be a nonempty Zariski open subset. Let $K_{X'}$ be the field defined above. Then:*

1. *$K_{X'} \mid \mathcal{M}(X)$ is a Galois extension.*
2. *For each base point $x \in X'$ there is a canonical isomorphism of profinite groups*

$$\mathrm{Gal}(K_{X'} \mid \mathcal{M}(X)) \;\cong\; \widehat{\pi_1(X', x)}.$$

Proof. (1) To see that $K_{X'}$ is Galois over $\mathcal{M}(X)$ it is enough to check that it is stable under the action of $\mathrm{Gal}(\overline{\mathcal{M}(X)} \mid \mathcal{M}(X))$. If $L \mid \mathcal{M}(X)$ comes from a compact Riemann surface and restricts to a cover over X', then any Galois conjugate $\sigma(L)$ is again a finite extension of $\mathcal{M}(X)$ of the same degree. Moreover, the property of being unramified above X' is invariant under conjugation, hence $\sigma(L)$ also appears in the defining composition. Thus $\sigma(K_{X'}) = K_{X'}$.

(2) We compare finite quotients on both sides.

Let G be a finite quotient of $\widehat{\pi_1(X', x)}$. Equivalently, G is a finite quotient of $\pi_1(X', x)$. Such a quotient corresponds to a connected finite Galois cover $p^{'} : Y^{'} \to X'$ with Galois group G. By the extension theorem for covers of punctured compact Riemann surfaces, $p^{'}$ extends uniquely to a finite Galois *branched* cover $p : Y \to X$. Then $\mathcal{M}(Y) \mid \mathcal{M}(X)$ is a finite Galois extension with group G, and by construction p is unramified over X', so $\mathcal{M}(Y) \subset K_{X'}$.

Conversely, let $L \mid \mathcal{M}(X)$ be a finite Galois subextension of $K_{X'} \mid \mathcal{M}(X)$ with group G. By Lemma 3.3, L comes from a finite holomorphic map $p : Y \to X$ restricting to a cover over X'. Over X' this gives a finite Galois topological cover $Y^{'} \to X'$ with group G. Hence G is a finite quotient of $\pi_1(X', x)$, and therefore of $\widehat{\pi_1(X', x)}$.

Finally, these correspondences are compatible with morphisms between finite quotients (equivalently, with towers of covers and towers of field extensions), so passing to inverse limits yields an isomorphism $\mathrm{Gal}(K_{X'} \mid \mathcal{M}(X)) \cong \widehat{\pi_1(X', x)}$. □

Remark 3.5. Let X be a connected compact Riemann surface. Every nonconstant holomorphic map $Y \to X$ is unramified over some Zariski open subset $X' \subset X$ (remove the finite branch locus). Hence every finite subextension of $\overline{\mathcal{M}(X)} \mid \mathcal{M}(X)$ is contained in some field $K_{X'}$ attached to a suitable X'. Furthermore, if we have $X' \subset X''$, then $K_{X'} \supset K_{X''}$. Therefore $\{K_{X'}\}_{X-X' \text{ is finite}}$ forms an inverse system, and we have

$$\operatorname{Gal}(\overline{\mathcal{M}(X)} \mid \mathcal{M}(X)) = \varprojlim_{X'} \operatorname{Gal}(K_{X'} \mid \mathcal{M}(X)) \tag{3.1}$$

$$= \varprojlim_{X'} \widehat{\pi_1(X', x)} \tag{3.2}$$

Example. Now take $X = \mathbb{P}^1(\mathbb{C})$, so $\mathcal{M}(X) \cong \mathbb{C}(t)$. For a finite set $\{x_1, \dots, x_n\} \subset \mathbb{P}^1(\mathbb{C})$ and a base point $x \notin \{x_1, \dots, x_n\}$, topology gives a presentation

$$\pi_1\big(\mathbb{P}^1(\mathbb{C}) \setminus \{x_1, \dots, x_n\}, x\big) = \langle \gamma_1, \dots, \gamma_n \mid \gamma_1 \cdots \gamma_n = 1 \rangle,$$

hence for $n > 1$ this group is isomorphic to the free group F_{n-1} on $n-1$ generators.

Corollary 3.6. *Every finite group occurs as the Galois group of some finite Galois extension $L \mid \mathbb{C}(t)$.*

Proof. Let G be a finite group, generated by (say) m elements. Choose $n := m + 1$ and pick distinct points $x_1, \dots, x_n \in \mathbb{P}^1(\mathbb{C})$. Then $\pi_1(\mathbb{P}^1(\mathbb{C}) \setminus \{x_1, \dots, x_n\}, x) \cong F_{n-1} = F_m$ admits a surjection onto G.

Passing to profinite completions gives a surjection $\widehat{\pi_1(\mathbb{P}^1(\mathbb{C}) \setminus \{x_1, \dots, x_n\}, x)} \twoheadrightarrow G$. By Theorem 3.4 (applied with $X' = \mathbb{P}^1(\mathbb{C}) \setminus (\{x_1, \dots, x_n\} \cup \{\infty\})$ or by simply removing the finitely many punctures) the group $\widehat{\pi_1(\mathbb{P}^1(\mathbb{C}) \setminus \{x_1, \dots, x_n\}, x)}$ is a finite quotient of $\operatorname{Gal}(\overline{\mathbb{C}(t)} \mid \mathbb{C}(t))$. Thus G is a finite quotient of $\operatorname{Gal}(\overline{\mathbb{C}(t)} \mid \mathbb{C}(t))$, i.e. the Galois group of some finite Galois extension $L \mid \mathbb{C}(t)$. □

4. The absolute Galois group of $\mathbb{C}(t)$

The corollary describes the finite quotients of $\operatorname{Gal}(\overline{\mathbb{C}(t)} \mid \mathbb{C}(t))$, but not the group itself. The right structure is a profinite analogue of a free group.

4.1. **Free profinite groups.** Let **Grp** denote the category of (discrete) groups and **ProfGrp** the category of profinite groups with continuous homomorphisms. There is a forgetful functor

$$U : \mathbf{ProfGrp} \longrightarrow \mathbf{Set},$$

sending a profinite group to its underlying set.

A first idea is to define a *free profinite group on a set X* as a left adjoint to this forgetful functor. That is, one would like a profinite group F together with a map $i : X \to U(F)$ such that for every profinite group G and every map $\lambda : X \to U(G)$, there exists a unique continuous homomorphism $\varphi : F \to G$ making the diagram

$$\begin{array}{ccc} X & \xrightarrow{\lambda} & U(G) \\ {\scriptstyle i}\downarrow & \nearrow_{U(\varphi)} & \\ U(F) & & \end{array}$$

commute.

However, it turns out that this is *impossible*. In other words, the forgetful functor U : **ProfGrp** $\to$ **Set** does not admit a left adjoint. The obstruction comes from the topology: in a profinite group, continuity of homomorphisms imposes strong finiteness conditions. This leads to the following modification of the universal property:

Let X be a set. Instead of considering all maps $X \to U(G)$, we restrict attention to those maps that are compatible with the profinite topology.

Definition 4.1. A profinite group F together with a map $i : X \to F$ is called a *free profinite group on* X if it satisfies the following property:

for every profinite group G and every map $\lambda : X \to G$ such that for every open normal subgroup $N \triangleleft G$ all but finitely many elements of $\lambda(X)$ lie in N, there exists a unique continuous homomorphism

$$\varphi : F \to G$$

with $\lambda = \varphi \circ i$.

Construction of the free profinite group. Let X be a set and let $F(X)$ denote the free (discrete) group on X. Consider the family of normal subgroups $N \triangleleft F(X)$ of finite index with the property that N contains all but finitely many elements of X. This family is directed under inclusion.

Definition 4.2. The *free profinite group on* X, denoted $\widehat{F}(X)$, is defined as the inverse limit

$$\widehat{F}(X) \;:=\; \varprojlim_{N} F(X)/N,$$

where N ranges over normal subgroups of finite index in $F(X)$ containing all but finitely many elements of X.

By construction, $\widehat{F}(X)$ is a profinite group. The canonical map $X \to F(X) \to \widehat{F}(X)$ defines a map $i : X \to \widehat{F}(X)$.

Proposition 4.3. *The pair* $(\widehat{F}(X), i)$ *satisfies the modified universal property of a free profinite group on* X.

Proof. Let G be a profinite group and let $\lambda : X \to G$ be a map such that for every open normal subgroup $N \triangleleft G$ all but finitely many elements of $\lambda(X)$ lie in N. By the universal property of the free discrete group $F(X)$, there exists a unique group homomorphism

$$\lambda_0 : F(X) \to G$$

extending λ.

The given finiteness condition implies that λ_0 is continuous when $F(X)$ is endowed with the profinite topology defined above. Hence λ_0 factors uniquely through the profinite completion $\widehat{F}(X)$, yielding a continuous homomorphism $\varphi : \widehat{F}(X) \to G$ with $\lambda = \varphi \circ i$. Uniqueness follows from the density of $i(X)$ in $\widehat{F}(X)$. □

4.2. **Douady's Theorem.**

Theorem 4.4. *There is an isomorphism of profinite groups*

$$\mathrm{Gal}(\overline{\mathbb{C}(t)} \mid \mathbb{C}(t)) \;\cong\; \widehat{F}(\mathbb{C}),$$

i.e. the absolute Galois group of $\mathbb{C}(t)$ *is the free profinite group on the set* $\mathbb{C}$.

Proof. For a finite subset $S \subset \mathbb{C}$, set

$$X_S := \mathbb{P}^1(\mathbb{C}) \setminus \big(S \cup \{\infty\}\big), \qquad \text{and fix a base point } x_0 \in X_S.$$

Let $K_S := K_{X_S}$. By Theorem 3.4, we have a canonical isomorphism

$$\mathrm{Gal}\big(K_S \mid \mathbb{C}(t)\big) \;\cong\; \widehat{\pi_1(X_S, x_0)}.$$

Since X_S is a sphere with $|S|+1$ punctures, its fundamental group is a free group of rank $\mathrm{card}(S)$, hence

$$\widehat{\pi_1(X_S, x_0)} \;\cong\; \widehat{F}(S).$$

We choose, for each S, an identification

$$\mathrm{Gal}\big(K_S \mid \mathbb{C}(t)\big) \;\cong\; \widehat{F}(S)$$

sending the standard small loops around points of S to the basis elements of S.

If $S \subset T$ are finite, then $X_T \subset X_S$, hence $K_S \subset K_T$ inside $\overline{\mathbb{C}(t)}$. By Galois theory there is a restriction map

$$\rho_{ST} : \mathrm{Gal}\big(K_T \mid \mathbb{C}(t)\big) \twoheadrightarrow \mathrm{Gal}\big(K_S \mid \mathbb{C}(t)\big).$$

Topologically, the inclusion $X_T \hookrightarrow X_S$ kills the loops around the points in $T \setminus S$, so under the identifications above, ρ_{ST} corresponds to the canonical surjection

$$\widehat{F}(T) \twoheadrightarrow \widehat{F}(S) \quad \text{that sends } x \mapsto 1 \text{ for } x \in T \setminus S.$$

Now the directed union of the fields K_S (over all finite $S \subset \mathbb{C}$) is $\overline{\mathbb{C}(t)}$. Therefore, by infinite Galois theory,

$$\mathrm{Gal}\big(\overline{\mathbb{C}(t)} \mid \mathbb{C}(t)\big) \;\cong\; \varprojlim_S \mathrm{Gal}\big(K_S \mid \mathbb{C}(t)\big) \cong \varprojlim_S \widehat{F}(S) \cong \widehat{F}(\mathbb{C}).$$

The last equality is very intuitive. However, it needs some arguments. It follows from the proposition below:

Proposition 4.5. *Let X be a set and $\mathcal{S}$ the directed system of finite subsets $S \subset X$. For $S \subset T$, let*

$$\lambda_{ST} : \widehat{F}(T) \twoheadrightarrow \widehat{F}(S)$$

be the canonical continuous epimorphism sending each generator $x \in T \setminus S$ to 1. Then

$$\varprojlim_{S \in \mathcal{S}} \widehat{F}(S) \;\cong\; \widehat{F}(X).$$

Sketch of proof. We list three lemmas and then explain how they combine to prove the proposition.

Lemma 4.6. *Let (G_S, λ_{ST}) be an inverse system of profinite groups over $\mathcal{S}$. For each $x \in X$, the family of elements*

$$g_{x,S} \in G_S \qquad (S \in \mathcal{S},\ x \in S)$$

with the compatibility condition $\lambda_{ST}(g_{x,T}) = g_{x,S}$ for all $S \subset T$ defines a unique element $g_x \in \varprojlim_S G_S$. Moreover, specifying the elements g_x for all $x \in X$ is equivalent to specifying such a compatible family.

Lemma 4.7. *Let $G := \varprojlim_S \widehat{F}(S)$ with transition maps killing $T \setminus S$. Let $i : X \to G$ be the map induced by the canonical maps $S \to \widehat{F}(S)$ and Lemma 4.6. Then (G, i) satisfies the modified universal property of the free profinite group on X: for any profinite group H and any map $\lambda : X \to H$ such that for every open normal subgroup $N \triangleleft H$ all but finitely many $\lambda(x)$ lie in N, there exists a unique continuous homomorphism $\varphi : G \to H$ with $\lambda = \varphi \circ i$.*

Lemma 4.8. *Let G be a profinite group and let $A \subset G$ be a subset whose closure is G. If $\varphi_1, \varphi_2 : G \to H$ are continuous homomorphisms into a Hausdorff topological group H and $\varphi_1|_A = \varphi_2|_A$, then $\varphi_1 = \varphi_2$.*

How the lemmas are used. We apply Lemma 4.6 to the system $G_S = \widehat{F}(S)$: for each $x \in X$, and each finite $S \ni x$, we have the distinguished generator $x \in \widehat{F}(S)$. The compatibility of transition maps implies these generators form a compatible family, hence define an element $i(x) \in G := \varprojlim_S \widehat{F}(S)$. This gives the map $i : X \to G$.

Now let H be a profinite group and let $\lambda : X \to H$ satisfy the finiteness condition in the modified universal property. For each finite $S \subset X$, the restriction $\lambda|_S : S \to H$ extends uniquely to a continuous homomorphism

$$\varphi_S : \widehat{F}(S) \to H$$

by the universal property of $\widehat{F}(S)$ (here S is finite, so it is just the usual profinite completion of a free group). Moreover, if $S \subset T$, then φ_T and $\varphi_S \circ \lambda_{ST}$ agree on the generators in S, and φ_T kills the generators in $T \setminus S$ because $\lambda(x)$ is required to be trivial modulo arbitrarily small open normal subgroups for all but finitely many x; hence the family $(\varphi_S)_S$ is compatible. Therefore it defines a unique continuous homomorphism

$$\varphi : G = \varprojlim_S \widehat{F}(S) \to H$$

with $\lambda = \varphi \circ i$. This proves existence in Lemma 4.7. Uniqueness follows from Lemma 4.8, because the closed subgroup generated by $i(X)$ is dense in G (indeed, its projection to each $\widehat{F}(S)$ contains the generators, hence is all of $\widehat{F}(S)$).

Thus (G, i) satisfies the same modified universal property as $\widehat{F}(X)$. By the usual Yoneda-type argument (universal objects are unique up to unique isomorphism), we conclude $G \cong \widehat{F}(X)$, as claimed. □

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